Introduction to the ab initio nuclear many-body problem

Lecture 1: *One- two- and many-body correlations*

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1 Introduction to many body correlations

Our aim will be to solve the many-body Schröndiger's equation from first principles, that is, without any assumptions or uncontrolled approximations. We'll work both in first and second quantization of quantum mechanics so maybe the most appropriate point of departure is establishing these two languages for the many fermion problem and the connection between them

The wavefunction of N particles obeys the many-body Schröndiger's equation

$$i\hbar\partial_t\Psi(\mathbf{x}_1,\mathbf{x}_2,\ldots,\mathbf{x}_N;t) = H\Psi(\mathbf{x}_1,\mathbf{x}_2,\ldots,\mathbf{x}_N;t)\ . \tag{1}$$

As a rule we'll use lower case letters (ψ, ϕ, ...) for the wavefunctions of a single particle and capital letters (Ψ, Φ, ...) wavefunctions of many particles. The many-body Hamiltonian is *many-body* operator (see below for what exactly that means) and in coordinate space it takes the form

$$H = \sum_{i=1}^{N} T(\mathbf{x}_i) + \sum_{i<j}^{N} V_2(\mathbf{x}_i,\mathbf{x}_j) + \sum_{i<j<k}^{N} V_3(\mathbf{x}_i,\mathbf{x}_j,\mathbf{x}_k) + \cdots\ , \tag{2}$$

where T is the kinetic energy operator (also to be specified clearly in a bit), and V_2, V_3, etc., are two-, three-, etc., particle potentials. Contrary to many introduction to many-particle quantum mechanics in condensed matter theory, here we explicitly allow for many-particle forces. As we will see in Lecture 3, this is an essential element in the accurate description of many-nucleon systems.

The many-body wavefunction $\Psi(\mathbf{x}_1,\mathbf{x}_2,\ldots,\mathbf{x}_N)$ is a function of $d\times N$ coordinates and identifying (or inferring) its exact form is a central aim of *ab initio* nuclear theory. As with every-multivariate function we can expand the many-body wavefunction in sums of products of a complete set of functions that depend on fewer variables (as long as they provide the same asymptotic behaviours). As the simplest choice, we can choose a complete set of single-particle wavefunctions which will allow us an interpretation of each term in the sum as a configuration of particles in individual single-particle states. So we make a choice of a complete set $\{\phi_E(\mathbf{x})\}$ spanned by all possible values of E which stands for the necessary set of quantum numbers (e.g., $\mathbf{k}, s_z$ for fermions in a box) and the wavefunction becomes

$$\Psi(\mathbf{x}_1,\mathbf{x}_2,\ldots,\mathbf{x}_N;t) = \sum_{E_1 E_2\cdots E_N} C(E_1,E_2,\ldots,E_N;t)\phi_{E_1}(\mathbf{x}_1)\phi_{E_2}(\mathbf{x}_2)\cdots\phi_{E_N}(\mathbf{x}_N)\ . \tag{3}$$

Note that now the time-dependence is prescribed by the expansion coefficients C and that the wavefunction's normalization carries over to C as

$$\int d^dx_1 d^dx_2\cdots d^dx_N|\Psi(\mathbf{x}_1,\mathbf{x}_2,\ldots,\mathbf{x}_N;t)|^2 = \sum_{E_1 E_2\cdots E_N}|C(E_1,E_2,\ldots,E_N)|^2 = 1 \tag{4}$$

Of course, the choice of $\{\phi_E(\mathbf{x})\}$ is far from unique and that is a good thing: any complete set of single-particle states will do, but some will do better than others. The number of terms in Eq. (3) depends heavily on the choice of the single-particle states used. A sum of many terms is harder to interpret than a sum of few, even when a single coordinate is involved, e.g., $\sum_{n=1}^{1000}(-1)^n x^{2n+1}/(2n+1)!$ has an approximate period of 2π, but that's hard to tell by inspection. Even before any attempts for interpretation, the number of terms in Eq. (3) will determine if such an approach is even practical for calculations. As such one would like to choose the set of single-particle orbitals wisely, to reduce the number of terms entering Eq. (3) and provide practical methods. Finding the expansion coefficients C would mean having the many-body wavefunction, but doing this in a brute force is hopeless. To see this explicitly we can attempt it and plug Eq. (3) into the many-body Schröndiger's equation in Eq. (1) and projecting on one set of quantum numbers(multiplying with the hermitian and integrating of all $\mathbf{x}$'s) yielding

$$
\begin{aligned}
&i\hbar\partial_t C(E_1, E_2, \ldots, E_N; t) = \sum_{i=1}^{N} \sum_{W} \left[\int d^d x_i \phi^{\dagger}_{E_i}(x_i) T(x_i) \phi_W(x_i) \right] C(E_1, \ldots, E_{i-1}, W, E_{i+1}, \ldots, E_N; t) \\
&+ \sum_{i<j} \sum_{WW'} \left[\int d^d x_i d^d x_j \phi^{\dagger}_{E_i}(x_i) \phi^{\dagger}_{E_j}(x_j) V_2(\mathbf{x}_i, \mathbf{x}_j) \phi_W(x_i) \phi_{W'}(\mathbf{x}_j) \right] C(E_1, \ldots, E_{i-1}, W, E_{i+1}, \ldots, E_{j-1}, W', E_{j+1}, \ldots, E_N; t) \\
&+ \sum_{i<j<k} \sum_{WW'W''} \left[\int d^d x_i d^d x_j d^d x_k \phi^{\dagger}_{E_i}(x_i) \phi^{\dagger}_{E_j}(x_j) \phi^{\dagger}_{E_k}(x_k) V_3(\mathbf{x}_i, \mathbf{x}_j) \phi_W(x_i) \phi_{W'}(\mathbf{x}_j) \phi_{W'}(\mathbf{x}_k) \right] \times \\
&\qquad \times C(E_1, \ldots, E_{i-1}, W, E_{i+1}, \ldots, E_{j-1}, W', E_{j+1}, \ldots, E_{k-1}, W'', E_{k+1}, \ldots, E_N; t) \ . \qquad (5)
\end{aligned}
$$

For each set of quantum numbers $\{E_1, \ldots, E_N\}$, Eq. (5) yields a set of differential equations. Even when truncated, this approach is hopeless for most practical cases.

Note that up to this point we haven't mentioned anything about the particles' statistics, that is, the terms in the sum in Eq. (3) have some symmetries stemming from the indistinguishability of quantum particles. For fermions this manifests as a property of the wavefunction

$$\Psi(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_i, \ldots, \mathbf{x}_j, \mathbf{x}_N; t) = -\Psi(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_j, \ldots, \mathbf{x}_i, \mathbf{x}_N; t) \ , \qquad (6)$$

where for bosons RHS's sign would be positive. This property transfers to the expansion coefficients C as

$$C(E_1, E_2, \ldots, E_i, \ldots, E_j, E_N; t) = -C(E_1, E_2, \ldots, E_j, \ldots, E_i, E_N; t) \ . \qquad (7)$$

From this the Pauli exclusion principle follows which states that if two quantum numbers are the same, e.g., $E_1 = E_2$, then C vanishes. That is, in each configuration that describes the many-body wavefunction, no two particles can be in the same state. As we'll see later this introduces the lowest rank of correlations to the system because each particle knows some information about what other particles are doing. More importantly the indistinguishability of particles also allows for a more convenient description in terms of occupation numbers, that is, we can describe each configuration by the number of particles that occupy each state in the set $\{E\}$. Forgetting for the moment that only one fermion at a time can occupy state each E, we can define the coefficients

$$\bar{C}(n_1, n_2, \ldots, n_\infty) = C(E_1, E_1, \overset{\cdots}{n_1}, E_2, E_2, \overset{\cdots}{n_2} \cdots, E_2, \ldots) \ , \qquad (8)$$

Note that a single permutation of the arguments of C on the LHS would just introduce a minus sign. We could now parse all terms in Eq. (3) and replace all instances of $C(E_1, E_1, \overset{\cdots}{n_1}, E_2, E_2, \overset{\cdots}{n_2} \cdots, E_2, \ldots)$ and its permutations multiplied by the appropriate minus signs. We would be making $N!/n_1!n_2! \cdots n_\infty!$ replacements of a C coefficient with the *same* $\bar{C}$ coefficient times a sign. This means that in Eq. (4), every $N!/n_1!n_2! \cdots n_\infty!$ terms will be the same:

$$1 = \sum_{E_1 E_2 \cdots E_N} |C(E_1, E_2, \ldots, E_N)|^2 = \sum_{n_1 n_2 \cdots n_\infty} |\bar{C}(n_1, n_2, \ldots, n_\infty)|^2 \frac{N!}{n_1! n_2! \cdots n_\infty!} \qquad (9)$$

This tells of a symmetry which we have already specified: the wavefunction is fully antisymmetric under the exchange of particles. To exploit this symmetry we define two things. First, we define the coefficients f

$$f(n_1, n_2, \ldots, n_\infty) = \left(\frac{N!}{n_1! n_2! \cdots n_\infty!} \right)^{\frac{1}{2}} \bar{C}(n_1, n_2, \ldots, n_\infty) \qquad (10)$$

such that $\sum_{n_1 \cdots n_\infty} |f(n_1, \ldots, n_\infty|^2 = 1$. Second, we examine each replacement we made realizing that a minus sign was introduced only when the C coefficient that was replaced was an odd permutation of particle indices away from the one in Eq. (8). This means that if we group all terms that correspond to a set of occupation numbers $n_1, n_2, \ldots, n_\infty$, we can define a many-particle wavefunction

$$\Phi_{n_1 n_2 \cdots n_\infty}(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N) = \left(\frac{n_1! n_2! \cdots n_\infty!}{N!}\right)^{\frac{1}{2}} \sum_{\sigma \in \mathcal{P}_N} (-1)^\sigma \phi_{E_1}(\mathbf{x}_\sigma(1)) \phi_{E_2}(\mathbf{x}_\sigma(2)) \cdots \phi_{E_N}(\mathbf{x}_\sigma(N)) \tag{11}$$

$$= \left(\frac{n_1! n_2! \cdots n_\infty!}{N!}\right)^{\frac{1}{2}} \begin{vmatrix} \phi_{E_1}(\mathbf{x}_1) & \phi_{E_1}(\mathbf{x}_2) & \cdots & \phi_{E_1}(\mathbf{x}_N) \\ \phi_{E_2}(\mathbf{x}_1) & \phi_{E_2}(\mathbf{x}_2) & \cdots & \phi_{E_2}(\mathbf{x}_N) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{E_N}(\mathbf{x}_1) & \phi_{E_N}(\mathbf{x}_2) & \cdots & \phi_{E_N}(\mathbf{x}_N) \end{vmatrix} , \tag{12}$$

where we have introduced the inverse of the combinatorics factor we discussed before for later convenience, σ is any permutation in the permutation group $\mathcal{P}_N$ that contains all permutations of N objects and $(-1)^\sigma$ is positive (negative) for even (odd) permutations. For the second line of Eq. (12), we recognized the definition of a determinant. This object is called a *Slater determinant* and we'll list its properties further below. For now we'll only note that they are a complete basis of antisymmetric functions of many-variables. Putting all definitions together, we can finally write Eq. (3) in terms of the occupation numbers,

$$\Psi(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N) = \sum_{n_1 n_2 \cdots n_\infty}^{1} f(n_1, n_2, \ldots, n_\infty) \Phi_{n_1, n_2, \ldots, n_\infty}(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N) \ . \tag{13}$$

This is the form of the wavefunction in second quantization where we use occupation numbers instead of sets of quantum numbers and each term in the expansion is explicitly antisymmetric.

In order to take full advantage of the simplification that second quantization has to offer, one also defines the state vector

$$|n_1 n_2 \cdots n_\infty\rangle \tag{14}$$

such that

$$\langle \mathbf{x}_1 \mathbf{x}_2 \cdots \mathbf{x}_N | n_1 n_2 \cdots n_\infty \rangle = \Phi_{n_1 n_2 \cdots n_\infty}(\mathbf{x}_1, \mathbf{x}_2, \ldots \mathbf{x}_N) \tag{15}$$

and corresponding creation and annihilation operators, $a_i, a_i^\dagger$ such that

$$a_i^\dagger |n_1 n_2 \cdots n_i \cdots n_\infty\rangle = \sqrt{n_i + 1} \, |n_1 n_2 \cdots n_i + 1 \cdots n_\infty\rangle \ , \tag{16}$$

$$a_i |n_1 n_2 \cdots n_i \cdots n_\infty\rangle = \sqrt{n_i} \, |n_1 n_2 \cdots n_i - 1 \cdots n_\infty\rangle \ , \tag{17}$$

which obey the *anti-commutation* relations

$$\{a_i, a_j\} = 0 \tag{18}$$

$$\{a_i, a_j^\dagger\} = \delta_{ij} \ . \tag{19}$$

Note that this means that the creation and annihilation operators carry with them (in their algebra) the commutation relation such that one doesn't have to worry about them. Further, $\langle \mathbf{x} | a_i^\dagger |0\rangle = \phi_i(\mathbf{x})$, that is, each creation or annihilation operator carries with it a single-particle wavefunction. One then finds that each state vector can be constructed as

$$\left(a_1^\dagger\right)^{n_1} \left(a_2^\dagger\right)^{n_2} \cdots \left(a_\infty^\dagger\right)^{n_\infty} |0\rangle = \prod_{i=1}^{\infty} \left(a_i^\dagger\right)^{n_i} |0\rangle = |n_1 n_2 \cdots n_\infty\rangle \ , \tag{20}$$

where

$$|0\rangle = |00\cdots 0\rangle \tag{21}$$

is the vacuum state. Putting all these definitions together and with some manipulation one arrives at the second quantized version of the Hamiltonian

$$H = \sum_{ij} \langle i| T |j\rangle a_i^\dagger a_j + \frac{1}{2}\sum_{ijkl} \langle ij| V_2 |kl\rangle a_i^\dagger a_j^\dagger a_l a_k + \frac{1}{3!}\sum_{ijklmn} \langle ijk| V_3 |lmn\rangle a_i^\dagger a_j^\dagger a_k^\dagger a_n a_m a_l + \cdots \tag{22}$$

where note the reverse ordering of the annihilation operators. We have come full circle back to the many-body Hamiltonian but now in second quantization. This definition can be extended now to any other operator $n - body$ operator A

$$A(\mathbf{x}_{i_1}, \mathbf{x}_{i_2}, \ldots, \mathbf{x}_{i_N}) = \sum_{i_1 \neq i_2 \neq \cdots i_n = 1}^{N} A(\mathbf{x}_{i_1}, \mathbf{x}_{i_2}, \ldots, \mathbf{x}_{i_n}) = \sum_{\substack{k_1 k_2 \cdots k_n \\ l_1 l_2 \cdots l_n}} \langle k_1 k_2 \cdots k_n| A |l_1 l_2 \cdots l_n\rangle a_{k_1}^\dagger a_{k_2}^\dagger \cdots a_{k_n}^\dagger a_{l_n} \cdots a_{l_2} a_{l_1} \ , \tag{23}$$

where now the definition of a n-body operator is clear: when acting on a state $|n_1 n_2 \cdots n_\infty\rangle$ it redistributes n occupation numbers.

2 The Hartree-Fock method: getting the most out of a single Slater Determinant

In the previous section we saw how the Slater Determinant can act as a basis for the many-body wavefunction. Here we will see how far a single element of that basis can take us when trying to describe the ground-state of a system. That is we will fomulate the method that provides the Slater Determinant (SD) with the lowest energy. This is the Hartree-Fock(HF) method.

When applying the HF method to the ground state, one attempts to identify the SD with the lowest energy. Since the SD uniquely defines the structure of the state, what is left to be identified are the single-particle states that compose it (see Eq. (12). In other words, one needs to identify an appropriate single-particle basis; we will use the letter ϕ to symbolize the entire basis. (The choice of words here is not accidental: as it turns out, the basis ϕ is often considered ideal for building a multiple SD description after the fact.) As such, we are looking for the state $|\mathrm{HF}(\phi)\rangle$ such that

$$E_{\mathrm{HF}(\phi)} = \langle \mathrm{HF}(\phi)| H |\mathrm{HF}(\phi)\rangle \ , \tag{24}$$

with

$$|\mathrm{HF}(\phi)\rangle = \prod_{i=1}^{N} a_i^\dagger |0\rangle \ , \qquad \text{with} \quad \langle \mathbf{x}| a_i^\dagger |0\rangle = \phi_i(\mathbf{x}) \ . \tag{25}$$

Here we assumed an ordering of the basis ϕ such that the first N states are occupied in the HF state. These states we will call *hole states* as they can only be occupied by holes, which are, by definition the absence of particles. The states with $i > N$ we will call *particle states* as they can only be occupied by particles. Somewhat confusingly, in the HF state, the particle (hole) states are completely occupied by holes (particles) (and so they are unoccupied by particles (holes)).

The wavefunctions of the type in Eq. (25) are also sometimes called *Hartree products* in quantum chemistry, because before antisymmetrization they are a product of single-particle orbitals, and in these states the probability density of finding particles 1,2,...,N within volume elements $d\mathbf{x}_1$,$d\mathbf{x}_2$,...,$d\mathbf{x}_N$ centered at positions $\mathbf{x}_1$,$\mathbf{x}_2$,...,$\mathbf{x}_N$, is a product of probabilities,

$$|\phi_1(\mathbf{x}_1)|^2\, d\mathbf{x}_1\, |\phi_2(\mathbf{x}_2)|^2\, d\mathbf{x}_2 \cdots |\phi_N(\mathbf{x}_N)|^2\, d\mathbf{x}_N \tag{26}$$

Formulating the problem this way might make one suspect that it could be recast as a single-particle problem. This is mostly the case, as what we are after are single particle orbitals. However, for the SD to minimize the many-particle Hamiltonian, each single-particle orbital must have some information about the state of the other particles. In the previous section we showed how an SD arises as the description of a many-particle system where each particle moves in a predefined orbital, not explicitly of the existence of other particles (e.g., the orbital $\phi_1(\mathbf{x})$ take in the coordinates of only one particle). Hence, in our problem the information about the other particles must come from the definition of the orbitals. We will seek to construct these orbitals, i.e., single-particle basis, as the eigenfunctions of an one-body operator h (this guarantees orthogonality and the orbitals will be a basis):

$$h(\mathbf{x})\phi_i(\mathbf{x}) = \epsilon_i \phi_i(\mathbf{x}) \ . \tag{27}$$

As we saw in the previous section, a one-body operator (that now acts on the *many-body* wavefunction) can be constructed from h as,

$$H_{\mathrm{HF}} = \sum_{i=1}^{N} h(\mathbf{x}_i) \ . \tag{28}$$

As the subscript suggests, this one-body operator uniquely defines the HF state and it often goes by the name of *Hartree-Fock potential*. The problem is now recast as almost a single-particle one: we need to find h and its connection to the full many-body Hamiltonian H. The plan now is the following: we will express the energy $E_{\mathrm{HF}}(\phi)$ in Eq. (24) in a convenient formulation and we will show how its minimization over all possible bases ϕ defines h from the many-body Hamiltonian H.

The efficient formulation mentioned above is the one of the density matrices which we will briefly define now. Our basis ϕ is unknown, however, we can express it in terms of a known basis χ as

$$\phi_i = \sum_j D_{ji} \chi_j \ . \tag{29}$$

where D is a unitary transformation that connects on complete and orthogonal basis to the other and so $DD^\dagger = 1$. We can associate the basis χ with a set of creation (annihilation) operators $c_i^\dagger(c_i)$, in the sense that $\langle\mathbf{x}|\, c_i^\dagger \,|0\rangle = \chi_i(\mathbf{x})$ leading to the relation between operators

$$a_i^+ = \sum_j D_{ji} c_j^\dagger \ . \tag{30}$$

The density matrix of the HF state on the basis ϕ is defined a

$$\rho_{ij} = \langle \mathrm{HF}(\phi)|\, a_j^\dagger a_i \,|\mathrm{HF}(\phi)\rangle = \delta_{ij}\theta(N-i) \ , \tag{31}$$

that is, the density matrix of a SD on its own single-particle basis is diagonal with 1's for $i \leq N$ and 0's for $i > N$. This is an alternative definition of a SD. Because of this, diagonalizing a density matrix cannot provide much information about the unoccupied states which have eigenvalue 0. Furthermore, any unitary transformation that doesn't mix occupied ($i \leq N$) states and unoccupied ones ($i > N$) leaves the corresponding SD unchanged (up to an overall phase). A consequence of Eq. (31) is that $\mathrm{Tr}\rho = N$. Another consequence of Eq. (31) is that for a SD, $\rho^2 = \rho$. This is another definition of a SD.

One can then define the density matrix of the HF state, i.e., the SD composed of the basis ϕ, on the basis χ,

$$\rho_{ij} = \langle \mathrm{HF}(\phi)| c_j^\dagger c_i |\mathrm{HF}(\phi)\rangle = \sum_{kl} D_{ik} D^\star_{jl} \langle \mathrm{HF}(\phi)| a_k^\dagger a_l |\mathrm{HF}(\phi)\rangle = \sum_{i=k}^{N} D_{ik} D^\star_{jk} \ . \tag{32}$$

We can now slightly reformulate the problem: we seek the state with density matrix ρ that yields minimum energy (this refers to the full many-body Hamiltonian H) in the set of states with

$$\mathrm{Tr}\rho = N \quad \text{and} \quad \rho^2 = \rho \ . \tag{33}$$

We will now express E_{MF} using the HF state's density matrix and demand that it minimizes it by requiring that any small variation in ρ should increase the energy.

We will restrict ourselves to two-body forces but it should be straightforward how any of these expressions generalizes to higher-body forces. Our Hamiltonian are the first two terms in Eq. (2), which we write in the χ basis (i.e., exchanging a's for c's)

$$H = \sum_{ij} t_{ij} c_i^\dagger c_j + \frac{1}{4} \sum_{ijkl} \bar{v}_{ij,kl} c_i^\dagger c_j^\dagger c_l c_k \tag{34}$$

where we have used the shorthand $t_{ij} = \langle i| T |j\rangle$ and we have also taken advantage of the antisymmetry of the two-particle wavefunctions, as is very often done, and written the potential matrix elements as

$$\bar{v}_{ij,kl} = v_{ijkl} - v_{ijlk} \ . \tag{35}$$

Deriving the form of E_{HF} ifn Eq. (24) in terms of the density matrix is an exercise in Wick's theorem and it yields

$$E_{\mathrm{HF}}(\phi) = \sum_{ij} t_{ij} \rho_{ij}(\phi) + \frac{1}{2} \sum_{ijkl} \bar{v}_{ij,kl} \rho_{ki}(\phi) \rho_{lj}(\phi) = \mathrm{Tr}\left[t\rho(\phi)\right] + \frac{1}{2} \mathrm{Tr}_1 \mathrm{Tr}_1 \left[\rho(\phi) \bar{v} \rho(\phi)\right] \ . \tag{36}$$

where Tr_1 is an obvious shorthand notation. Following the plan we'll now investigate what a small variation in ρ does to $E_{\mathrm{HF}}[\rho]$. Following the calculus of variations, changing ρ by $\delta\rho$ into $\rho' = \rho + \delta\rho$ results in a change in δE_{HF} in E_{HF},

$$\delta E_{\mathrm{HF}} = E_{\mathrm{HF}}[\rho] - E_{\mathrm{HF}}[\rho - \delta\rho] = \sum_{ij} \frac{\partial E_{\mathrm{HF}}[\rho]}{\partial \rho_{ij}} \delta\rho_{ij} \ , \tag{37}$$

where we've only kept terms linear in $\delta\rho$. We would like δE_{HF} to vanish for any variation in ρ (up to the constraints in Eq. (33), but we will ignore this for now, keep the variation general and address this constraint below.) As it turns out the single-particle Hamiltonian we were looking for in Eq. (27) is $h_{ij} = \partial E_{\mathrm{HF}}[\rho]/\partial\rho_{ij}$ and from Eq. (36) we get

$$h_{ij} = \partial E_{\mathrm{HF}}[\rho]/\partial\rho_{ij} = t_{ij} + \Gamma_{ij} \ , \tag{38}$$

where

$$\Gamma_{ij} = \sum_{kl} \bar{v}_{il,jk} \rho_{kl} \ , \tag{39}$$

is the so-called *Fock operator*. At this point we'll have to consider the constraints of Eq. (33) and especially the second one (the first one is relatively simple to enforce in practice.): the variations in ρ should happen such that ρ remains the density matrix of a SD, or equivalently,

$$(\rho + \delta\rho)^2 = \rho + \delta\rho \tag{40}$$

and keeping up to linear terms in $\delta\rho$,

$$\delta\rho = \rho\delta\rho + \delta\rho\rho \tag{41}$$

In the HF basis, where ρ is diagonal and acts a projector on the occupied states (because its 0 anywhere else) this means that

$$\rho\delta\rho\rho = 0 = (1-\rho)\delta\rho(1-\rho) \tag{42}$$

or in words, to make sure that we are considering variations within the subset of many-particle states that are SDs, we must only make variations that don't change the particle-particle (pp;ρ_{ij}, with $i, j \leq N$) and hole-hole (hh;ρ_{ij}, with $i, j > N$) parts of the density matrix, in the HF-basis. That is, in the HF basis, we can only allow for variations of the particle-hole (ph) or hole-particle (ph) parts of the density matrix. Which provides the constraint

$$[h, \rho] = 0 \tag{43}$$

for $h = t + \Gamma$. Note one could start from Eq. (43) immediately as it dictates that h from Eq. (38) and ρ should be simultaneously diagonalizable which means that ρ is an SD made of the eigenfunctions of h. In more detail, since the diagnoalization of ρ is determined up to a unitary transformation within particle states or hole states, we can choose this transformation to be the one that diagonbalizes h and so we are left with an eigenproblem of h,

$$hD = \epsilon D \tag{44}$$

$$\sum_k h_{ik}D_{kj} = \sum_k \left(t_{ij} + \sum_{l=1}^{N}\sum_{pr} \bar{v}_{ir,kp} D_{pl} D^{\star}_{ri} \right) D_{kj} = \epsilon_j D_{ij} \ . \tag{45}$$

This equation determines D which determines ϕ via Eq. (29). We are now done: we have derived a one-body Hamiltonian,

$$H_{\rm HF} = \sum_{i=1}^{N} h(\mathbf{x}_i) = \sum_{ij} h_{ij} c_i^\dagger c_j = \sum_i \epsilon_i a_i^\dagger a_i \ , \tag{46}$$

where ϵ_i is often called the Hartree-Fock spectrum. The ground state of this Hamiltonian is the HF state, a SD composed of the N lowest eiegnefunctions of h which yields the minimum energy for the full many-body Hamiltonian of all other SD states. The Hartree-Fock energy, that is, the energy of the full many-body Hamiltonian at the Hartree-Fock state is then

$$E_{\rm HF} = \sum_{i=1}^{N} \epsilon_i - \frac{1}{2}\sum_{ij=1}^{N} \bar{v}_{ij,ij} \ . \tag{47}$$

Before we finish, let's see explicitly why Hartree-Fock is a mean-field method. If we write Eq. (45) in coordinare space we get

$$\left(-\frac{\hbar^2}{2m}\boldsymbol{\nabla}^2 + \Gamma_H(\mathbf{x}) \right) \phi_k(\mathbf{x}) + \int d\mathbf{x}' \Gamma_F(\mathbf{x}, \mathbf{x}') \phi(\mathbf{x}') = \epsilon_k \phi_k(\mathbf{x}) \tag{48}$$

where

$$\Gamma_H(\mathbf{x}) = \int d\mathbf{x}' v(\mathbf{x}, \mathbf{x}')\rho(\mathbf{x}') \tag{49}$$

$$\Gamma_F(\mathbf{x}) = -v(\mathbf{x}, \mathbf{x}')\rho(\mathbf{x}, \mathbf{x}') \tag{50}$$